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Transforming Hydraulic Fracturing with Automation: Field Results and Impact on Operational Consistency and Efficiency

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Abstract

Automation has become increasingly crucial for enhancing efficiency in unconventional resource development, specifically in hydraulic fracturing applications, representing a significant paradigm shift in completions practices. The implementation of automated fracturing provides a pathway to standardize operations, minimize variability, enhance safety, and optimize resource utilization. This paper outlines the implementation and field evaluation of automated fracture design execution systems, focusing on their impact on operational performance and exploring potential for future advancements. Automated fracturing systems were deployed to execute the hydraulic fracturing design with minimal human intervention, maintaining real-time, precise control over treatment parameters such as rate, pressure, and proppant concentration. The methodology involved system calibration, operational workflow redesign, and validation through real-time monitoring of treatment quality and screen-out risk detection with programable responses to reduce associated risk. Field results indicate significant improvements in stage execution efficiency, treatment consistency, and decisions made throughout the treatment. The automated systems reduced cycle times, improved adherence to fracture design parameters, and enhanced responsiveness to downhole events, such as early sign of screen-out, compared to conventional manual operations. Observations also demonstrated a reduction in human error, improved safety outcomes, and higher consistency throughout the operational phase. One of the most notable outcomes was the increase in proppant ratio performance, a variable that compares the design to actual proppant pumped in a hydraulically-fractured stage, nearly reaching 99%. The results confirm that automation in fracturing execution delivers measurable operational benefits. Automated fracturing represents a critical evolution in completion practices, establishing a new standard for unconventional well stimulation and delivery. The novelty of this work lies in improving the operational efficiency ofhydraulic fracturing operations. Automated fracturing introduces new capabilities such as real-time data analytics enhancement, which will act asthe foundation for fully autonomous well completions. This paper will review the results of this technology and its impact on stimulation overall efficiency.

Introduction

In recent years, the global exploration and exploitation of unconventional hydrocarbon resources have witnessed a remarkable surge. This trend is primarily driven by escalating global energy demands, which have prompted the industry to explore increasingly complex and challenging unconventional resources. The successful development of these resources has been made possible by significant technological advancements, particularly in horizontal drilling and hydraulic fracturing. These techniques have unlocked vast reserves previously considered inaccessible, leading to a revolution in energy production. The evolution of the industry has been paralleled by innovations in fracturing technologies and methodologies, driving improved treatment outcomes and increased operational efficiency. Today, automation is emerging as a pivotal enabler in the next phase of this evolution, transforming fracturing operations from labor-intensive, reactive processes into highly controlled, data-driven workflows. This shift is not only improving operational performance, but also redefining how operators approach safety, consistency, and economic viability in completion design and execution. The increasing complexity of multi-stage fracturing in horizontal wells has placed new demands on operational precision. With large number of stages being executed across extended lateral lengths, even minor inconsistencies in treatment parameters (e.g., injection rate, treatment pressure, or proppant concentration) can lead to significant variations in fracture geometry, and ultimately, production performance. Traditional manual control methods, which are effective in simpler scenarios, struggle to maintain the level of repeatability and responsiveness required in today's high-density, pad-based development environments. Automation addresses these challenges by introducing a level of control and adaptability that manual operations cannot match. Through the integration of programmable logic controllers (PLCs), supervisory control and data acquisition systems, and real-time analytics, automation enables continuous monitoring and dynamic adjustment of fracturing parameters. This ensures that each stage is executed according to itsengineering design, with minimal deviation and maximum efficiency. Moreover, automation supports the implementation of standardized operating procedures across crews and locations, reducing variability and enhancing overall campaign performance. Beyond operational efficiency, automation also plays a critical role in enhancing safety and risk management. This is evident in remote operations that reduces the need for personnel to be in close proximity to high-pressure equipment Additionally, predictive algorithms and anomaly detection systems allow for early identification of potential issues, for example, pressure build-ups or proppant bridging, enabling proactive intervention before they escalate into screen-outs. This paper presents a comprehensive methodology for integrating automation into hydraulic fracturing operations, focusing on key phases from system calibration and initiation to execution, real-time monitoring, and contingency planning. It explores how automation enhances control over critical treatment parameters, improves consistency across stages and wells, and supports data-driven decision-making. This paper demonstrates the tangible benefits of automation in real-life application examples highlighting its impact on operational efficiency and execution. It aims to provide operators, completion engineers, and operational representatives with actionable insights into the implementation and optimization of automated fracturing systems in today's dynamic and data-rich operational landscape.

Background

The evolution of hydraulic fracturing from a manually intensive operation to a highly automated process reflects the broader digital transformation sweeping across the upstream oil and gas industry. As unconventional resourcesbecome more complex, the need for precision, repeatability, and real-time adaptability has never been greater. Notably, the operational setup where multiple wells are placed in the same well pad, further amplifies the importance of efficiency. With multiple wells requiring sequential or simultaneous operations, even minor delays or inconsistencies can have a compounding effect on the overall execution timeline. Automation addresses these challenges by integrating advanced control systems, and data analyticsinto the fracturing workflow, enabling operators to achieve superior performance with reduced risk. A key enabler for transitioning to a fully automated approach in this case is the clear and comprehensive understanding of the source rock formation, established through prior exploration, development, and production activities. These insights allow for a smooth shift from manual decision-making to parameter-driven automated execution, as formationbehavior, fracture response, and operational constraints are already well understood. Historical performance data from previous wells provides a robust foundation for defining optimal treatment parameters and automation logic. This well history and prior analysis have led to notable improvements across several operational aspects, including optimization of clean fluid volumes, enhanced frac initiation practices, and more efficient proppant placement strategies. Such adjustmentsare the direct result of lessons learned from past stimulation jobs and their evaluation, allowing for more accurate stage designs, further reducing variability between treatments. The benefits of this knowledge base extend into automation by enabling tighter operational control and faster decision-making when adjustments are required. At its core, automation enhances operational consistency by minimizing human intervention and its associated variability. In traditional fracturing operations, decisions are often made based on the experience of operators with real-time judgment, which introduces inconsistencies across stages, between crews and operators. Automated systems, on the other hand, execute fracturing treatments based on predefined parameters and real-time response, ensuring that each stage is performed with the same level of precision, regardless of location, crew, or time of day. Moreover, automation significantly improves safety by reducing the need for personnel to be in high-risk zones during fracturing operations. Remote monitoring and control capabilities allow operators to manage the treatment and pumping of hydraulic fracturing stages from a centralized control room, minimizing exposure to high-pressure equipment and hazardous environments. This not only enhances personnel safety but also supports compliance engineering design. From a technical standpoint, automation enables tighter control over critical fracturing parameters, which are interdependent and must be carefully managed to achieve optimal fracture geometry and uniformity. Automated control systems use closed-loop feedback mechanisms to dynamically adjust pump rates, proppant feeders, and fluid blending units in response to real-time downhole conditions. This level of precision is difficult to achieve manually and is essential for ensuring placement uniformity of all stages, which ultimately lead to optimum reservoir contact. Another key advantage of automation lies in its ability to integrate with advanced data analytics and machine learning platforms. Automation systems not only process this data in real time but also store it for post-job evaluation and reference. These sets of data can establish a strong foundation for applying machine learning algorithms to identify trends, predict equipment failures, and optimize future treatments. This data-driven approach supports continuous improvement and enables operators to move from reactive to predictive maintenance strategies. Economically, automation drives significant cost savings by reducing non-productive time (NPT), limiting equipment wear, and enhancing overall operational efficiency. Automated systems can identify and respond to anomalies more quickly than human operators, helping to prevent costly screen-outs, equipment failures, and treatment deviations. Furthermore, their ability to execute treatments with greater speed and precision shortens operational durations, potentially reducing total completion costs. The operational analysis in this paper is based on two wells, each located in a separate multi-well pad. One well was completed using the manually controlled fracturing method, and the other with an automated fracturing workflow. The wells share similar geological settings, completion designs, and surface infrastructure, enabling a direct, representative comparison of the impact of automation on efficiency, treatment quality, and performance.

Methodology

System Calibration

Prior to initiating the fracturing treatment, a comprehensive system calibration wasundertaken to ensure that all surface equipment, control systems, and measurement devices operate within specified tolerances, serving as a critical quality control step to minimize unplanned deviations from the engineering design during stimulation. The calibration sequence includes verification of high-pressure pump performance, covering rate capability andpressure integrity under anticipated loads, followed by evaluation of fluid and proppant delivery systems to confirm that blending equipment, metering devices, and conveyors can deliver consistent concentrations and rates as specified in the treatment schedule. Measurement instrumentation, including pressure transducers and flow meters, werecalibrated by zeroing and span-checking against certified reference gauges, verifying accuracy across the expected operating range, with results documented to establish baseline performance. Integrated automated diagnostics and system health checks were embedded within the data acquisition platform to validate sensor inputs, actuator responses, and communication integrity, flagging anomalies such as sensor drift, pressure inconsistencies, or communication failuresin real time, allowing corrective actions to be taken before pumping begins. This structured calibration and diagnostic process ensured that all controllable parameters remain consistentwith the treatment design, reduces the probability of screen-outs or rate fluctuations, and lowers the overall risk of operational delays or equipment failures.

Frac Initiation and Automation. The fracture initiation phase marks the beginning of the fracturing treatment, where communication with the formation is established through the controlled injection of fracturing fluid. Automation plays a pivotal role by precisely managing the initial injection rate and pressure ramp-up through closed-loop control systems that integrate real-time surface and calculated downhole pressure response. These systems dynamically adjust pump rates to maintain a smooth pressure build-up, accounting for factors such as fluid compressibility, perforation friction, and pressure derivative with rate. This precise control minimizes reaction time, reduces variability between stages, and limits the risk of premature fracture initiation. This resulted in a more consistent initiation process that improved stage-to-stage repeatability, enhanced fracture placement accuracy, and supportedthe overall treatment efficiency across the application well.

Controllable Parameters. In the pre-job setup, critical operational parameters are programmed into the control platform to ensure optimal fracture geometry, efficient proppant placement, and consistent stage execution. These parameters include the maximum allowable treating wellhead pressure, target pumping rate, stage pumping schedule with defined sand concentrations and pumping cycles, and the specified friction reducer (FR) concentration. Once configured, the automation system autonomously executes the stage sequence through regulating pump output, blender performance, additive injection, and proppant delivery in strict accordance with the treatment designwhile continuously monitoring real-time data to maintain operational accuracy. At any point during the treatment, the frac crewmaintainsthe ability to intervene or adjust parameters as necessary, ensuring flexibility to respond to unforeseen conditions. This parameter-driven approach standardizedexecutionacross all stages, enhanced treatment repeatability, and enabled adaptive optimization in real time, which wasparticularly advantageous in multi-well pad operations.

Proppant Introduction– Design Driven. The timing of proppant introduction is a critical decision point in the fracturing process. Introducing proppant too early can lead to near-wellbore screen-out, while delaying it can result in inadequate fracture extension. Automated systems use design-driven triggers based on pressure response and cumulative fluid volume to determine the optimal moment to begin proppant introduction. This eliminated subjective decision-making and ensured that proppant wasintroduced only after the fracture has been adequately extended.

Flush Setting – Mechanical Reading. The flush phase is essential for cleaning the wellbore and ensuring that all proppant is effectively placed within the fracture. Automatedsystems utilize mechanical readings from surface sensors to determine the optimal flush volume and duration accounting for surface frac lines volume. The system can dynamically adjust the flush schedule using real-time data to avoid under-flushing, which may leave residual proppant in the wellbore, or over-flushing, which can reduce near-wellbore fracture conductivity.

Remote Real-Time Monitoring

Remote real-time monitoring has become a cornerstone of fracturing automation. User-friendly platforms and secure data transmission enable operators to monitor key parameters such as pressure, injection rate, and proppant concentration from centralized onsite control rooms or offsite locations. This not only enhanced safety by reducing personnel exposure to high-pressure environments but also facilitated rapid decision-making based on real-time data. Additionally, pre-programmed visualization tools and dashboards provided actionable insights, allowing operators to respond proactively and adjust as needed.

Proactive Contingency Planning

Automation systems are equipped with predictive algorithms to detect anomalies and execute contingency plans before they escalate into operational failures. Screen-outs, one of the most common and costly challenges in hydraulic fracturing, is identified early through continuous analysis of pressure trends, rate fluctuations, and proppant concentration data. Machine learning models, trained on historical and real-time data, provide high-accuracy predictions of screen-out events, allowing the system to proactively adjust operational parameters or alert operators to initiate corrective actions before premature fracture bridgingoccurs. These models are continuously updated and refined as new data becomes available, enhancing their predictive precision withtime. The automation platform integrates closed-loop control, enabling real-time adjustments such as cut proppant and start early flush, reduce injection rates, or adjust pumping schedules to maintain fracture and wellbore integrity. Contingency protocols are pre-programmed and tested under various operational scenarios to ensure robustness and reliability.

Results and Discussion

Consistency in Pumping Rate

The analysis compares two wells from two separate pads; one completed using the standard(manual) pumping method and the other utilizing an automated fracturing system. Both wells underwent similar treatment designs and operational conditions, providing a basis for direct comparison. The two plots, as shown in Figure 1, display pumping rate profiles over the treatment duration for various fracturing stages. The first plot on the left, representing the manually controlled well, exhibited noticeable fluctuations in pumping rates between stages, reflecting the variability of manual adjustments and operator response times. Conversely, the second plot on the right, corresponding to the automated well, showed a much more consistent pumping rate, closely maintaining the target rate throughout most stages. Quantitatively, this difference is quantifiedin the coefficient of variation(C_v) (relative standard deviation) values as shown in Table 1, indicating greater variability for the standard method compared to the automated process. These results demonstrate that automation substantially improved operational consistency, which is critical for adhering tothe treatment design, ensuring a more uniform fracture geometry, and optimizing proppant placement, ultimately enhancing the overall effectiveness of the fracturing treatment and operational efficiency.

$$C_v = \frac{\sigma}{\mu}$$

Also,

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}\left(x_i - \mu\right)^2$$

Where:

C_v:Coefficient of variation
σ^2:Variance
μ:Arithmetic mean

Table 1—The coefficient of variation calculated for each technique (manual and automated)

Parameter	Manual Frac	Automated Frac
C_v	0.03	0.003

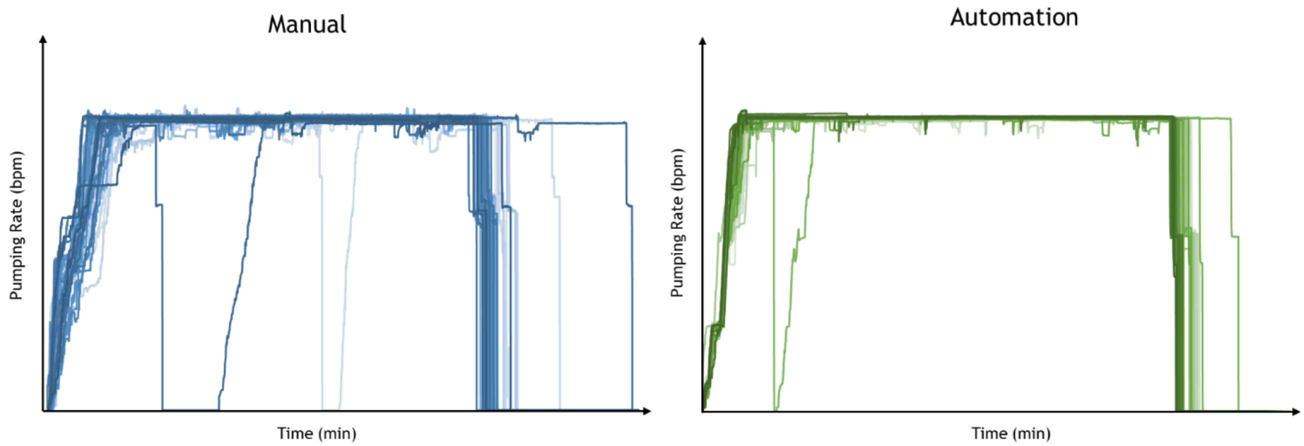

Figure 1—Consistency in pumping rate: (blue) manual frac treatment; (green) automated frac treatment

Rate Ramp-up

Figure 2 illustrates pumping rate profiles for a single fracturing stage from each well. The automated system reached the target pumping rate approximately 10 to15% faster during fracture initiation compared to the manual method. This quicker ramp-up improvedthe overall operational efficiency by 5 to 7%, significantly improving daily completion efficiency.

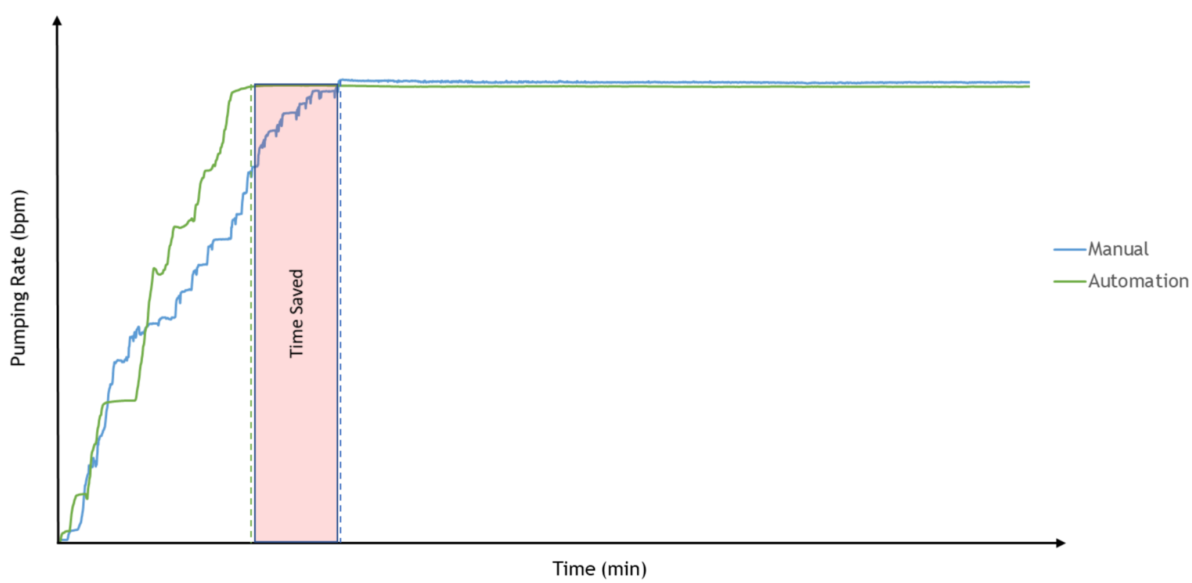

Figure 2—Rate ramping up during frac initiation stage: (blue) manual frac treatment; (green) automated frac treatment

Frac Execution Consistency

This section compares the average treating pressure against the average pumping rate across multiple fracturing stages for each well. As inferred by Figure 3, the data from the manually operated well exhibit greater scatter, reflecting higher variability in pressure and rate control between stages. In contrast, the automated well shows a tighter clustering of data points, indicating more consistent adherence to target operational parameters. This enhanced consistency, achieved with automation highlighted the improved process control and repeatability, which are critical for uniform fracture placementand optimizing treatment outcomes.

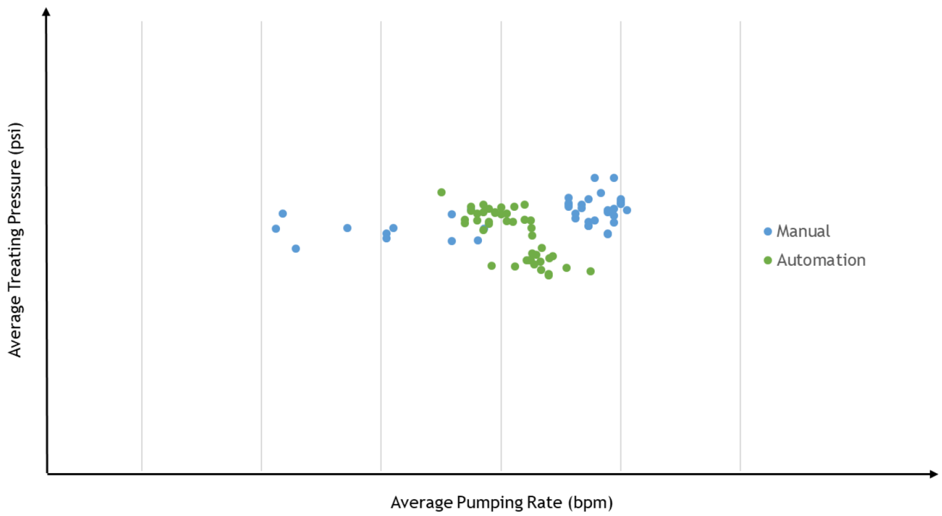

Figure 3—Frac execution consistency: (blue) manual frac treatment; (green) automated frac treatment

Proppant Concentration Accuracy

Proppant concentration accuracyis defined as the ratio of the designed proppant-to-fluid volume to the actual proppant-to-fluid volume delivered during treatment. The results shown in Figure 4 demonstrates that the automated system achieved a proppant concentration accuracy of 99%, compared to 95% for the manual operation. This highlights that automation significantly enhances the precision and consistency of proppant placement via a more consistent pumping schedule execution.

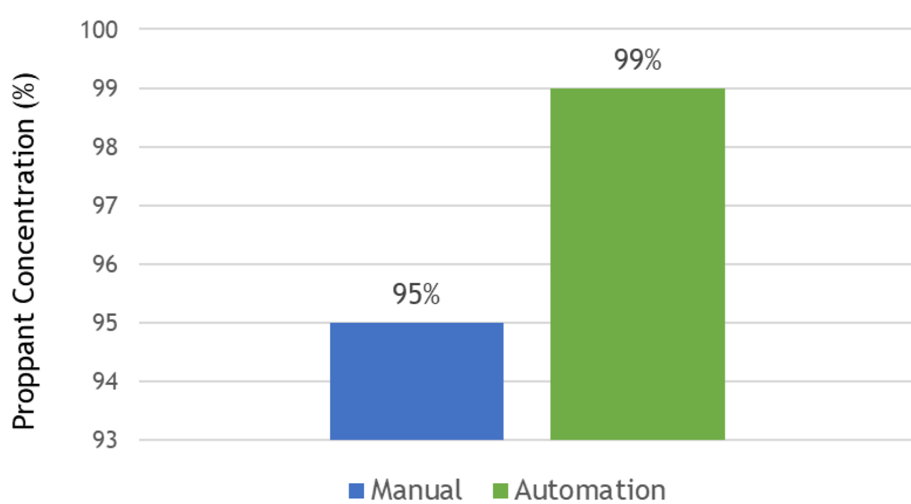

Figure 4—Proppant concentration percentage: (blue) manual frac treatment; (green) automated frac treatment

Excess Fluid Volume

The calculation of excess fluid volume revealed a notable difference between manual and automated operations, with the manual method exhibiting approximately 5% excess fluid volume, compared to around 1% with the automated system as highlighted in Figure 5. Furthermore, Figure 6 illustrates the cumulative volume of clean fluid injected over the entire treatment period for a representative stimulation stage in each well. This reduction in excess fluid not only conserves valuable water and chemicals but also enhances overall operational efficiency by freeing up capacity to accommodate additional pumping time. Consequently, automation contributed to more sustainable resource management.

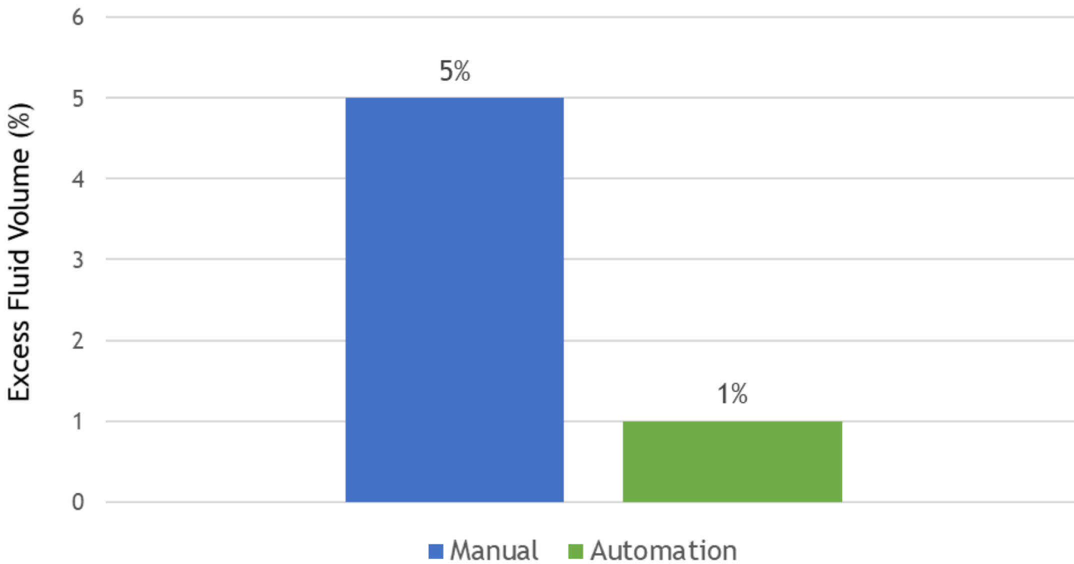

Figure 5—Excess fluid volume: (blue) manual frac treatment; (green) automated frac treatment.

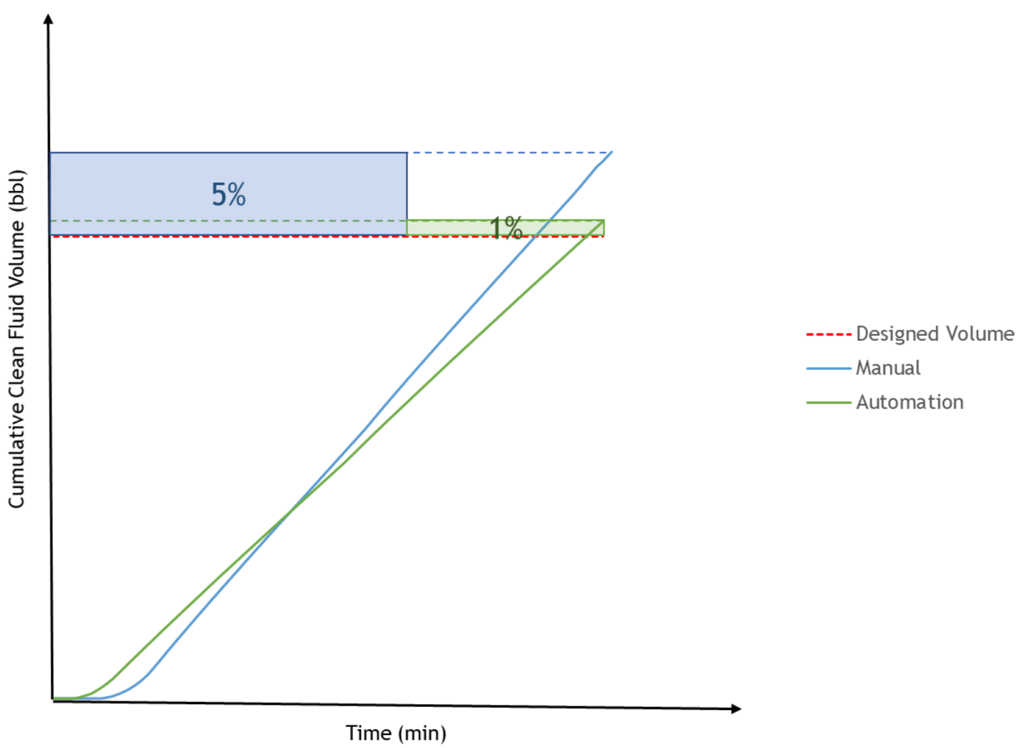

Figure 6—Cumulative clean fluid volume per stage: (blue) manual frac treatment; (green) automated frac treatment.

Conclusion

This paper demonstrated that automated hydraulic fracturing systems offer substantial improvements over traditional manual operations in unconventional resources. By comparing two wells under similar conditions, the automated approach showed enhanced consistency in critical parameters such as pumping rate, treating pressure, and proppant concentration. Notably, the coefficient of variation in pumping rate decreased by tenfolds from 0.03 in the manual operation to 0.003 with automation, highlighting superior consistencyand repeatability. Automation also supported execution standardization by embedding standard pumpingprocedures and schedules into the control system, guiding the fracturing crew through each step and minimizing human intervention. This ensures every stage is performed with precision and consistency, regardless of individual preference or location-specific factors, which is essential for maximizing efficiencyand reducing operational risk. The automation system also achieved a 10 to 15% faster ramp-up rate to target pumping rates, with an average savings of 7 minutes, improving operational efficiency by 5 to 7%, potentially accommodatingadditional pumping time per day. Proppant concentration accuracy reached 99% in the automated system compared to 95% manually, while excess fluid volume was reduced from 5% to 1%, reflecting better and more effective resource utilization. Beyond operational benefits, automation enhanced safety by reducing personnel exposure and supported proactive risk management through early anomaly detection. These findings confirm that automation not only improved fracturing quality and efficiency but also establishes a new standard of execution consistency, making it a critical advancement for the future of multi-well completions and fully autonomous operations.

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